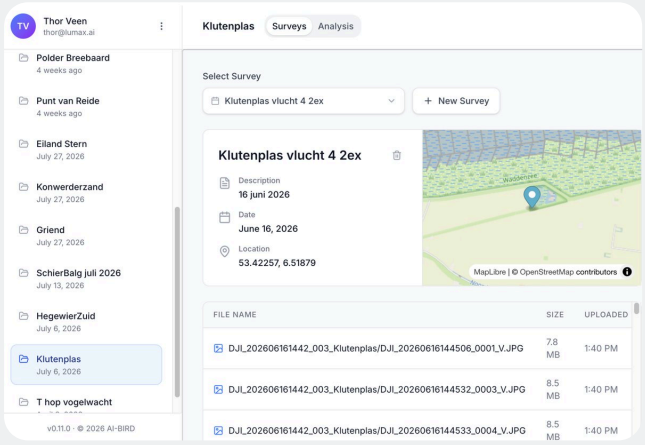
AI-BIRD

Automated Intelligence for Bird Inventory and Reproduction using Drones

Collecting numbers of breeding seabirds, colony development and breeding success is time intensive and causes disturbance, yet monitoring is increasingly important because of threats to populations due to e.g. avian influenza and El Niño. Aerial drone-based surveys in combination with AI allow for low-disturbance, fast, accurate and cost-efficient data collection.

AI-BIRD is a data analysis and storage platform that 1. turns drone imagery of seabird colonies into maps, 2. detects and classifies different bird species, their ages (adult, chick) and states (e.g. breeding, dead), and 3. presents the maps and results to the end-user through a web interface. Its unique aspects are user-friendliness and scalability: practitioners collect and upload drone data, optionally validate detections, and can access detection results and survey statistics within a short turnaround time, replacing days of manual field counting with a near real-time, repeatable process.



SURVEY TAB

Organise sites and flights, and upload geotagged drone imagery in a few clicks, no photogrammetry or GIS expertise or software needed.

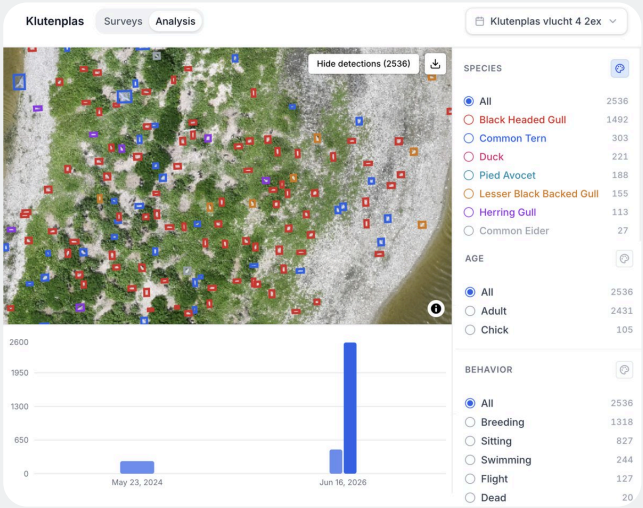
- Multiple sites and repeat surveys tracked in one place
- Flight metadata (date, location) captured automatically
- Raw drone imagery uploaded directly, no pre-processing required

ANALYSIS TAB*

Browse AI-detected birds on the orthorectified map, color-coded and filterable by species, age, and behaviour.

- Every detection classified and color-coded by species, age, and behaviour/state
- Customizable filters (e.g. one species, only chicks) to explore patterns instantly
- Count trends across repeat surveys shown alongside the map
- Option to download the detections

** Species list shortened in this preview to display the age and behaviour filters below it.*



WORKFLOW

Fly & upload
Drone survey collects geotagged photos, uploaded through the web app

→

Stitch
Photos combined into one georeferenced orthophoto using OpenDroneMap, no manual GIS work or software licenses needed

→

Detect & Classify
Two-stage AI model detects and classifies every bird

1 · Detect each bird

2 · Classify: species, age, state

→

Review (optional)
Ecologist verifies detections on the interactive map, correcting as needed

→

Analyse & monitor
Counts, locations & trends available for reporting across repeated surveys

MODEL PERFORMANCE

METRIC	DEFINITION	DETECTION	CLASSIFICATION	
			MEAN	SD
Precision	Proportion of all detections that were correct	**0.98**	**0.95**	0.06
Recall	Proportion of all birds present that were detected	**0.88**	**0.95**	0.04

Classification stats based on 10 species / 1,398 validated detections. Trained and validated on Dutch seabird colony surveys.

DATA PRODUCTS

- A georeferenced orthophoto (stitched aerial map) of the survey site
- Bird and nest detections and classifications (species, age, state) as a geo-referenced dataset
- An interactive web map for browsing, panning, and zooming the survey with detections overlaid (see example above)
- Survey statistics accessible on demand, enabling trend monitoring across surveys and sites over time

BUILT TO ADAPT

Model tuning: detection models can be retrained for site-specific requirements, covering new species or states not yet supported. **Scalable:** the platform is built to adapt to handle changing and rapidly increasing survey demands, such as a sudden avian flu outbreak. **Flexible engagement:** end-users can simply collect and upload data, or go further and provide data to train their own custom detection models. **Cloud-native:** data and app are hosted on Scaleway (a European cloud provider), so results can be shared instantly with anyone involved, no need to download or transfer files.

SUMMARY

AI-BIRD gives ecologists and conservation practitioners a **fast, cost-effective, and non-invasive** way to monitor seabird colonies and quickly detect die-offs due to e.g. avian flu. By automating the path from drone mapping to verified detection map, it turns a multi-day manual counting exercise into a same-day, repeatable digital workflow, making it practical to survey more colonies, more often, without added disturbance to the birds.

PARTNERS & FUNDING

AI-BIRD is developed by **Lumax** and **Sovon**, with financial support from the Dutch Ministry of Agriculture, Fisheries, Food Security and Nature (LVVN) and the EU's **NextGenerationEU** fund.

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